TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 1

(With Solutions)

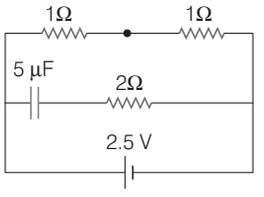
Duration : 3 Hours **Max. Marks : 720**

Instructions

- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- The question paper contains three parts of Physics, Chemistry and Biology respectively.
- Part A contains 45 Questions from Physics section. Part B contains 45 Questions from Chemistry section. Part C contains 90 Questions from Biology.

PART A PHYSICS

1. A capacitor of capacitance 5 μF is connected as shown in the figure. The internal resistance of the cell is 0.5 Ω. The amount of charge on the capacitor plates is



(a) 80 μC (b) 40 μC (c) 20 μC (d) 10 μC

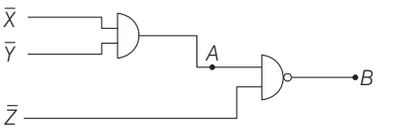
2. For a material the Young's modulus is 3.2 times that of rigidity modulus. Its Poisson's ratio is

(a) 0.6 (b) 0.3 (c) 0.2 (d) 0.5

3. In an L - C - R circuit inductance is changed from L to $L/2$. To keep the same resonance frequency, C should be changed to

(a) $2C$ (b) $\frac{C}{2}$ (c) $4C$ (d) $\frac{C}{4}$

4. Output B is given by



(a) $X + Y + Z$ (b) $\overline{XY} + Z$
(c) $\overline{X} + YZ$ (d) $X + XYZ$

5. A particle moving with uniform acceleration has average velocities v_1, v_2 and v_3 over the successive intervals of 5 s, 10 s and 15 s respectively. Then, the value of $\frac{v_1 - v_2}{v_2 - v_3}$ will be

(a) $\frac{5}{3}$ (b) 1 (c) $\frac{3}{5}$ (d) $\frac{1}{2}$

6. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale.

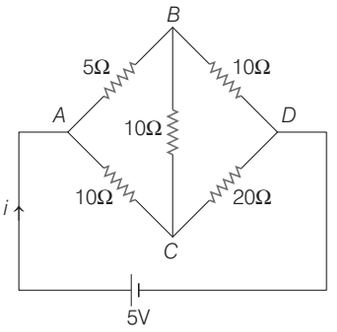
The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is

(a) 0.9% (b) 2.4% (c) 3.1% (d) 4.2%

7. An ideal solenoid having 5000 turns/m has an aluminium core and carries a current of 5 A. If $\chi_{Al} = 2.3 \times 10^{-5}$, then the magnetic field developed at centre will be

(a) 0.031 T (b) 0.048 T (c) 0.027 T (d) 0.050 T

8. Find current i in circuit shown in figure.



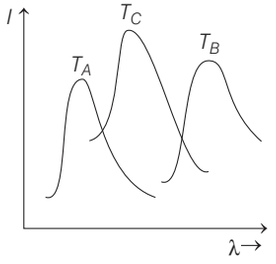
(a) 0.5 A (b) 0.2 A (c) 1 A (d) 2 A

9. A mass M is broken into two parts of masses m_1 and m_2. For the force of attraction or gravitation to be maximum between them, the m_1 and m_2 have value

(a) $m_1 = \frac{M}{3}, m_2 = \frac{2M}{3}$ (b) $m_1 = \frac{2M}{3}, m_2 = \frac{M}{3}$

(c) $m_1 = m_2 = \frac{M}{2}$ (d) $m_1 = \frac{M}{4}, m_2 = \frac{3M}{4}$

10. Kepler space telescope generated a plot of intensity of radiation *versus* wavelength of three stars in Andromeda galaxy. What is the relation between their temperature?



(a) $T_A > T_B > T_C$ (b) $T_C > T_B > T_A$

(c) $T_A > T_C > T_B$ (d) $T_B > T_C > T_A$

11. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5 s. In another 10 s it will decrease to α times its original magnitude, then α is equal to

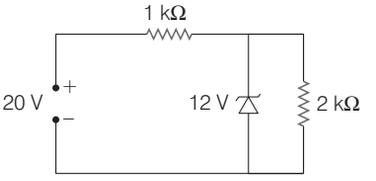
(a) 0.7 (b) 0.81 (c) 0.729 (d) 0.6

12. A heat engine placed on neptune (planet) operating between temperatures T_1 and T_2, where $T_1 > T_2$ has efficiency $\frac{1}{5}$. When T_2 is lowered by 60K its efficiency increase to $\frac{1}{2}$. Then, T_1 and T_2 are respectively

(a) 200 K and 180 K (b) 200 K and 160 K

(c) 252 K and 215 K (d) None of these

13. In the given circuit, the current through the resistor 2kΩ is



(a) 2 mA (b) 4 mA (c) 6 mA (d) 1 mA

14. The longitudinal waves starting from a ship return from the bottom of the sea to the ship after 2.60 s. If the bulk modulus of water be 220 kg mm^{-2} and the density is 1.1×10^3 kgm^{-3}, the depth of sea is (take, $g = 9.8$ N kg^{-1})

(a) 1820 m (b) 1850 m

(c) 1400 m (d) 1420 m

15. The maximum vertical distance through which a full dressed astronaut can jump on the earth is 0.5 m. The maximum vertical distance through which he can jump on the moon, which has a mean density (2/3) rd that of earth and radius one quarter that of earth is

(a) 1.5 m (b) 3 m (c) 6 m (d) 7.5 m

16. Two slits are separated by a distance of 0.5 mm and illuminated with light of λ = 6000 Å. If the screen is placed 2.5m from the slits, the distance of the third bright fringe from the centre will be

(a) 1.5 mm (b) 3 mm

(c) 6 mm (d) 9 mm

17. If λ_1 and λ_2 are the wavelengths of the first members of the Lyman and Paschen series respectively, then $\lambda_1 : \lambda_2$ is

(a) 1 : 3 (b) 1 : 30 (c) 7 : 50 (d) 7 : 108

18. The wavelength λ of a photon and the de-Broglie wavelength of an electron have the same value. Find the ratio of energy of photon to the kinetic energy of electron in terms of mass m, speed of light c and Planck constant h.

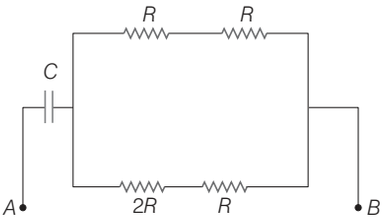
(a) $\frac{\lambda mc}{h}$ (b) $\frac{hmc}{\lambda}$

(c) $\frac{2hmc}{\lambda}$ (d) $\frac{2\lambda mc}{h}$

19. Potential energy of a particle in a force field is given by $U = \frac{A}{x^2} - \frac{B}{x}$ where, A and B are positive constant and x is distance of particle from centre of the field. For stable equilibrium, the distance of particle is

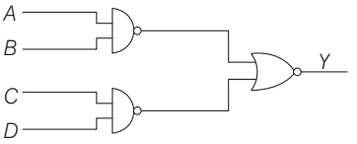
(a) $\frac{A}{B}$ (b) $\frac{B}{A}$ (c) $\frac{2A}{B}$ (d) $\frac{B}{2A}$

20. What will be the time constant for the given circuit?



(a) RC (b) $2.5\,RC$
(c) $1.2\,RC$ (d) $6\,RC$

21. The output Y of given logic circuit is



(a) $AB + CD$ (b) $ABCD$
(c) $ABC + D$ (d) $\overline{A \cdot B} + CD$

22. Match the corresponding entries of Column I with Column II and choose the correct option from the codes given below

	Column I		Column II
A.	Astigmatism	1.	Convex lens
B.	Hypermetropia	2.	Concave lens
C.	Myopia	3.	Cylindrical lens

Codes

	A	B	C		A	B	C
(a)	3	2	1	(b)	3	1	2
(c)	1	2	3	(d)	2	1	3

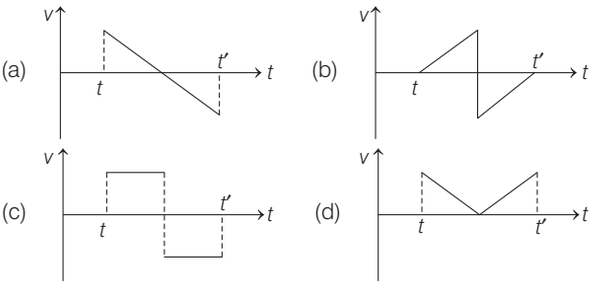
23. The effective resistance of n number of resistor when connected in parallel is $P\,\Omega$. If one of the resistor is removed, the effective resistance becomes $Q\,\Omega$. Then, the resistance of resistor which is removed is

(a) $\frac{PQ}{P+Q}$ (b) $\frac{PQ}{Q-P}$
(c) $\frac{P^2Q^2}{(Q-P)^2}$ (d) $\frac{PQ}{P-Q}$

24. The mass of jupiter is 1.9×10^{27} kg and that of sun is 1.99×10^{30} kg. The mean distance of the jupiter from the sun is 7.8×10^{11} m. Assume that the jupiter moves in a circular orbit around the sun. Velocity of jupiter is

(a) 1.3×10^4 ms^{-1} (b) 1.5×10^4 ms^{-1}
(c) 1.2×10^4 ms^{-1} (d) 1.6×10^4 ms^{-1}

25. In a match a sixer is hitted by a batsman and the ball touches the ground outside the boundary. Then, which of the following graph describes the variation of the cricket ball's vertical velocity v with time between the time t as it hits the bat and time t', when it touches the ground?



26. The excess pressure inside a spherical drop of water is five times that of another drop, then their respective mass ratio is

(a) 5 : 3 (b) 25 : 137
(c) 125 : 137 (d) 1 : 125

27. A thin symmetrical double convex lens of refractive index $\mu_2 = 1.5$ is placed between a medium of refractive index $\mu_1 = 1.4$ to the left and another medium of refractive index $\mu_3 = 1.6$ to the right. Then, the system behaves as

(a) a convex lens (b) a concave lens
(c) a glass plate (d) a convex or concave lens

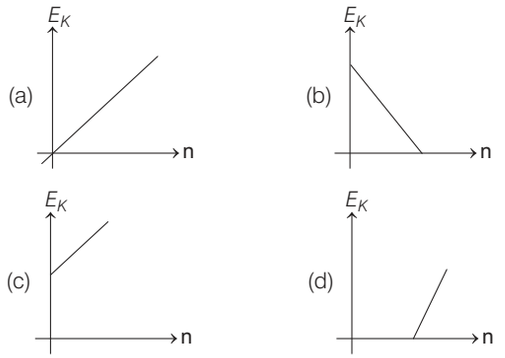
28. The half-life period of a radioactive element X is same as the mean life of another radioactive element Y. Initially, both of them have the same numbers of atoms, then

(a) X and Y have the same decay rate initially
(b) X and Y decay at the same rate always
(c) Y will decay at a faster rate than X
(d) X will decay at a faster rate than Y

29. At room temperature (27° C) a vehicle tyre has pressure of 3 atm. The temperature at which the tyre suddenly, burst will be (given, $\gamma_{air} = \frac{7}{5}$)

(a) 246K (b) 274K
(c) 217K (d) 204K

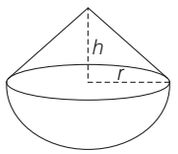
30. Which one of the following graph represents the variation of maximum kinetic energy E_K of the emitted electrons with frequency ν in photoelectric effect correctly ?



31. In Meldi's experiment a string vibrates in 3 loops, when 8 g were placed in the pan. To make the string vibrates in 5 loops, the mass should be placed is of value

(a) 2 g (b) 2.88 g
(c) 2.5 g (d) 4 g

32. A uniform solid right circular cone and a hemisphere of same base radius 2m and same density are placed on each other, so as to have common face. The centre of mass of the compositive solid lies on the common face. The height of the cone is



(a) 2m (b) 6m (c) 4m (d) 8 m

33. The resistance of the wire in platinum resistance thermometer at ice point is 3 Ω and at steam point is 9 Ω. When the thermometer is inserted in an unknown hot bath its resistance is found to be 12 Ω. The temperature of the hot bath is

(a) 100° C (b) 50° C (c) 75° C (d) 150° C

34. One mole of monoatomic gas and three moles of diatomic gas are put together in a container. The molar specific heat (in JK^{-1} mol^{-1}) at constant volume is (R = 8.3 JK^{-1} mol^{-1})

(a) 18.7 (b) 18.9 (c) 19.2 (d) 20.7

35. A body weighs 72 kgf on the surface of earth. Then, its weight on surface of mars, whose mass is $\frac{1}{9}$ and radius is $\frac{1}{2}$ of that of earth is

(here, kgf = kilogram-force)

(a) 72 kgf (b) 54 kgf (c) 62 kgf (d) 32 kgf

36. Two thin long parallel wires separated by a distance a are carrying a current I A each. The magnitude of the force per unit length will be

(a) $\frac{\mu_0 I}{4\pi a}$ (b) $\frac{\mu_0 I}{4\pi a^2}$ (c) $\frac{\mu_0 I^2}{2\pi a}$ (d) $\frac{\mu_0 I}{2\pi a}$

37. A billiard ball of mass m and radius r hit by a cue stick at a height h above the centre and it acquire a linear velocity v. The angular velocity ω acquired by the ball is

(a) $\frac{5vr^2}{2h}$ (b) $\frac{5vh}{2r^2}$ (c) $\frac{2vr^2}{5h}$ (d) $\frac{2vh}{5r^2}$

38. A current I flows in a conducting wire of length L. If we bent it in a circular form, its magnetic dipole moment would be

(a) $\frac{IL^2}{4\pi}$ (b) $\frac{IL}{4\pi}$ (c) $\frac{I^2L}{4\pi}$ (d) $\frac{I^2L^2}{4\pi}$

39. Minimum excitation potential of Bohr's first orbit in hydrogen atom is

(a) 3.6 V (b) 10.2 V (c) 13.6 V (d) 3.4 V

40. When a positive charge particle is moving on a equipotential surface, then work done in this process will be

(a) positive (b) negative (c) zero (d) None of these

41. Two plane mirror of length 24 m each are kept at a distance of 8 cm parallel to each other. A ray of light incident on one end of one mirror at angle of 53°. Then, total number of reflections before it reaches the other end is (given, $\tan 53° = \frac{4}{3}$)

(a) 225 (b) 175 (c) 250 (d) 200

42. A particle executes SHM of amplitude 25 cm and time period 3s. The minimum time required for the particle to move between two points 12.5 cm on either side of the mean position is

(a) 0.25 s (b) 0.5 s (c) 1 s (d) 1.25 s

43. A voltmeter has a resistance of G Ohm and range of V volt. The value of resistance used in series to convert it into a voltmeter of range nV volt is

(a) nG (b) $\frac{G}{(n-1)}$ (c) $(n-1)G$ (d) $\frac{G}{n}$

44. A source emits electromagnetic waves of wavelength 3m. One beam reaches the observer directly and other after reflection from a water surface, travelling 1.5 m extra distance with intensity reduced to 1/4 as compared to intensity due to the direct beam alone. The resultant intensity will be

(a) (1/4) fold (b) (3/4) fold (c) (5/4) fold (d) (9/4) fold

45. If 8 dipoles of charges of magnitude ± e are placed inside the cube. The electric flux coming out of cube will be

(a) $16e / E_0$ (b) $8e / E_0$ (c) e / E_0 (d) zero

PART B CHEMISTRY

46. The sulphide ore is converted to oxide before reduction because
(a) oxides are easier to reduce
(b) sulphide are easily decompose
(c) oxide get easily decompose
(d) sulphide get oxidised to sulphur

47. The separation of primary, secondary and tertiary amines can be done through fractional distillation. Which of the following method will not used to distinguish primary, secondary and tertiary amine?
(a) Hinsberg method
(b) Hofmann method
(c) Liebermann's nitroso method
(d) Victor Meyer's method

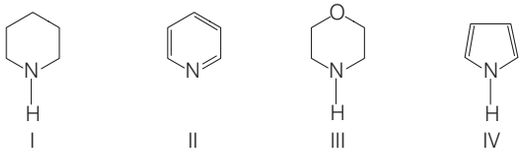
48. The amino acid which is optically inactive is
(a) alanine (b) lactic acid
(c) glycine (d) serine

49. The substance which is added to soap to impart antiseptic properties is
(a) iodoform (b) terpineol
(c) chloroxylenol (d) bithional

50. Strength of acid increases with the attachment of group showing $-I$ effect and decreases with the attachment of group showing $+I$ effect. Which of the following is correct sequence of basic strength in aqueous solution?
(a) $CH_3NH_2 < (CH_3)_2NH < (CH_3)_3N$
(b) $(CH_3)_2NH < CH_3NH_2 < (CH_3)_3N$
(c) $(CH_3)_3N < (CH_3)_2NH < CH_3NH_2$
(d) $(CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$

51. Compound $Ph-O-\overset{\overset{O}{\|}}{C}-Ph$ can be prepared by the reaction of
(a) phenol and benzaldehyde in the presence of palladium
(b) phenol and benzoic acid in the presence of NaOH
(c) phenol and benzoyl chloride in the presence of pyridine
(d) phenol and benzoyl chloride in presence of $ZnCl_2$

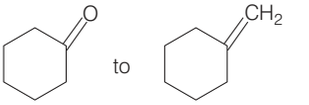
52. Arrange the following in correct sequence of basic strength.



I II III IV

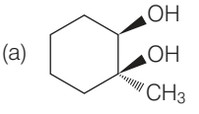
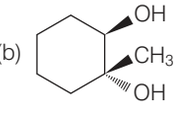
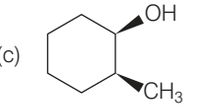
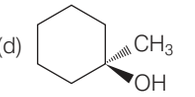
(a) IV > I > III > II (b) III > I > II > IV
(c) II > I > III > IV (d) I > III > II > IV

53. Which of the following reagent is used to convert?



(a) Zn-Hg in HCl (b) $PH_3P = CH_2$
(c) Both (a) and (b) (d) None of these

54. Product obtained on reaction of 2-methylcyclohexene with *m*-CPBA followed by hydrolysis is

(a) 
(b) 
(c) 
(d) 

55. One mole of the complex $CoCl_3 \cdot 6H_2O$ on reaction with excess of $AgNO_3$ gives two moles of white precipitate. Thus, complex is
(a) $[Co(H_2O)_6]Cl_3$ (b) $[Co(H_2O)_5Cl]\ Cl_2 \cdot H_2O$
(c) $[Co(H_2O)_4Cl_2]Cl \cdot 2H_2O$ (d) $[Co(H_2O)_3Cl_3] \cdot 3H_2O$

56. The ligand $N(CH_2CH_2NH_2)_3$ is
(a) tridentate (b) didentate
(c) tetradentate (d) pentadentate

57. The transition elements vary widely in their chemical behaviour. Many of them are sufficiently electropositive to dissolve in mineral acids, although a few are noble. Among the first series transition metals, except copper, all the metals are relatively reactive. Transition elements show variable valency due to
(a) small size
(b) large charge
(c) presence of incompletely filled *d*-subshell
(d) All of the above

58. The oxidation state of Ni in tetracarbonyl nickel is
(a) + 1 (b) + 2 (c) 0 (d) + 4

59. Chromatography is based on the principle of
(a) vaporisation (b) concentration
(c) absorption (d) adsorption

60. If the electron is visualised as a classical charged particle laterally rotating about an axis with angular momentum L and magnetic dipole moment M then find the magnetic moment of $K_3[FeF_6]$?
(a) 3.87 BM (b) 4.89 BM
(c) 5.91 BM (d) 6.92 BM

61. Which of the following inert gas element, shows the highest chemical reactivity?
(a) Xe (b) He
(c) Ne (d) Ar

62. The reduction of $>C=O$ to $>CH_2$ is done by Wolf-Kishner reduction. The reagent used in Wolf-Kishner reduction is
(a) NH_2—NH_2 in presence of base
(b) Zn—Hg in dil. HCl
(c) $PH_3P=CH_2$
(d) None of the above

63. The mixture of concentrated HCl and HNO_3 made in 3 : 1 ratio contains
(a) N_2O_4 (b) NOCl
(c) NCl_3 (d) ClO_2

64. Which of the following is/are use(s) of $KMnO_4$?
(a) In analytical chemistry
(b) As a favourite oxidant in preparative organic chemistry
(c) In bleaching of wool, cotton and silk
(d) All of the above

65. Which of the following statement is correct about salt bridge?
(a) Forms insoluble precipitate with the electrolytes in the two half-cells
(b) Enhances the flow of electrons by overcoming liquid junction potential
(c) Maintains the electrical neutrality of the two half-cells
(d) Provides the extra ions

66. Which of the following is not correct for the reaction?

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

(a) Rate of reaction is $-d[N_2O_5]$
(b) Rate of reaction is proportional to $[N_2O_5]^2$
(c) It is first order reaction
(d) It is redox reaction

67. The products formed on the photochemical decomposition of HNO_3 is
(a) N_2O_5 and NO_2 (b) N_2O_4 and NO_2
(c) NO_2 and O_2 (d) NO and O_2

68. Which of the following expression is correct for average kinetic energy of one molecule of an ideal gas at 27°C and 1 atm pressure?
(a) 900 cal K^{-1} $molecule^{-1}$
(b) 6.21×10^{-21} JK^{-1} $molecule^{-1}$
(c) 336.7 JK^{-1} $molecule^{-1}$
(d) 3741.3 JK^{-1} $molecule^{-1}$

69. Which of the following pair is isoelectronic?
(a) CO_2 and NO_2 (b) BF_3 and NH_3
(c) NO_3^- and CCl_4 (d) NO^+ and SbH_5

70. Match the block of elements given in Column I with their electronic configuration given in Column II and choose the correct code.

	Column I		Column II
A.	*s*-block	1.	[Xe] $4f^7 5d^1 6s^2$
B.	*p*-block	2.	[Kr] $4d^{10} 5s^1$
C.	*d*-block	3.	[Xe] $5s^1$
D.	*f*-block	4.	[Xe] $4f^{14} 5d^{10} 6s^2 6p^2$

Codes

	A	B	C	D
(a)	1	2	3	4
(c)	2	3	1	4

	A	B	C	D
(b)	3	4	2	1
(d)	3	4	1	2

71. The shortest wavelength transition in the Paschen series in hydrogen atom occurs at 821 nm. At what wavelength does it occur in Li^{2+} (isoelectronic to hydrogen)?
(a) 91.2 nm (b) 273.6 nm
(c) 821.0 nm (d) 7389.0 nm

72. Which of the following is the property of an isothermal expansion process?
(a) $\Delta U = 0$ (b) $W = -Q$
(c) $W = -nRT \ln \frac{V_2}{V_1}$ (d) All of these

73. Which of following statement is correct?
(a) Glucose has furanose structure and fructose has pyranose structure
(b) Glucose has pyranose structure and fructose has furanose structure
(c) Both have furanose structure
(d) Both have pyranose structure

74. A container has hydrogen and oxygen mixture in ratio of 1 : 1 by weight, then
(a) internal energy of the mixture decreases
(b) internal energy of the mixture increases
(c) entropy of the mixture increases
(d) entropy of the mixture decreases

75. In the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, α is that part of N_2O_4 which dissociates, then the number of moles at equilibrium will be
(a) $(1-\alpha)^2$ (b) 3α
(c) α (d) $1+\alpha$

76. The compound *A* on heating gives a colourless gas and a residue that is dissolved in water to obtain *B*. Excess of CO_2 is bubbled through aqueous solution of *B*, *C* is formed which recovered in the solid form. Solid *C* on gentle heating gives back *A*. The compound is
(a) $CaSO_4 \cdot 2H_2O$ (b) $CaCO_3$
(c) Na_2CO_3 (d) K_2CO_3

77. In Carius method, 0.099 g of organic compound gave 0.287 g of AgCl. The percentage of chlorine in the compound will be

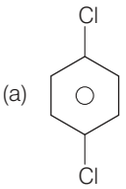
(a) 71.71% (b) 29.39%
(c) 53.06% (d) 76.23%

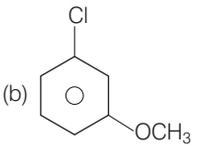
78. Choose the correct option which will fill the A, B and C correctly.

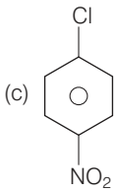
$$4B + 3CO_2 \longrightarrow A + C$$
$$4B + 3SiO_2 \longrightarrow B + Si$$
$$4B + 6NO \longrightarrow C + 3N_2$$

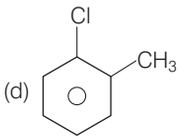
(a) B_2O_3, BO_2 and BO (b) B_2O_3, B_2O_2 and BO
(c) B_2O_3, B_2O_3 and B_2O_3 (d) BO, BO_2 and B_2O_3

79. Which of the following would react most readily with nucleophiles?

(a) 

(b) 

(c) 

(d) 

80. Match the following name of elements to stable end product.

	Name of elements		Stable end products
A.	Thorium	1.	$^{208}_{82}$ Pb
B.	Neptunium	2.	$^{209}_{83}$ Bi
C.	Uranium	3.	$^{206}_{82}$ Pb
D.	Actinium	4.	$^{207}_{82}$ Pb

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	4	1	(b)	3	4	1	2
(c)	1	2	3	4	(d)	4	1	3	2

81. Sewage containing organic waste should not be disposed in water bodies because it causes major water pollution. Fishes in such a polluted water die because of

(a) large number of mosquitoes
(b) increase in amount of dissolved oxygen
(c) decrease in amount of dissolved oxygen in water
(d) clogging of gills by mud

82. According to kinetic theory, the reactant molecules are assumed to be hard spheres and reaction is postulated to occur when molecules collide with each other. Pre-exponential factor in the expression obtained during collision theory is known as

(a) collision diameter (b) collision frequency factor
(c) collision energy (d) None of these

83. The number of nearest neighbours and next nearest neighbours of Na^+ ion in a crystal of NaCl are, respectively,

(a) $6Na^{\oplus}$, $12Cl^-$ (b) $6Cl^-$, $12Na^{\oplus}$
(c) 12 Cl^-, $12Na^{\oplus}$ (d) $6Cl^-$, $6Na^{\oplus}$

84. Each pair forms ideal solution except

(a) C_2H_5Br and C_2H_5I (b) C_6H_5Cl and C_6H_5Br
(c) C_6H_6 and $C_6H_5CH_3$ (d) C_2H_6I and C_2H_5OH

85. A certain current liberates 0.500 g of H_2 in 2.00 hr. How many gram of oxygen can be liberated by the same current in the same time?

(a) 0.500 g (b) 8.00 g
(c) 4.00 g (d) 16.00 g

86. Addition of 0.643 g of a compound to 50 mL of benzene (density = 0.879 g mL^{-1}), lowers the freezing point from 50.51°C to 50.03°C. If K_f for benzene is 5.12, the molecular mass (in g mol^{-1}) of the compound is

(a) 156.05 (b) 312.00
(c) 78.00 (d) 468.00

87. The oxygen atom in phenol

(a) exhibits only inductive effect
(b) exhibits only resonance effect
(c) has more dominating resonance effect than inductive effect
(d) has more dominating inductive effect than the resonance effect

88. At 500 K, the half-life period of a gaseous reaction at an initial pressure of 100 kPa is 364 s. When the pressure is 50 kPa, the half-life period is 182 s. The order of the reaction is

(a) zero (b) one (c) half (d) two

89. The effective atomic number of cobalt in $[Co(NH_3)_5H_2O]^{3+}$ is

(a) 36 (b) 33 (c) 24 (d) 30

90. Which of the following represents the correct order of solubility for the noble gases?

(a) Ne > Ar > Kr > He > Xe
(b) He > Ne > Ar > Kr > Xe
(c) Ar > Ne > He > Kr > Xe
(d) Xe > Kr > Ar > Ne > He

PART C BIOLOGY

91. The virus of hepatitis-A contains
(a) *ds*DNA (b) *ss*RNA
(c) *ss*DNA (d) *ds*RNA

92. Which of the following cranial nerves is incorrectly matched with its nature?
(a) Olfactory – Sensory
(b) Trochlear – Motor
(c) Glossopharyngeal – Mixed
(d) Hypoglossal – Sensory

93. Match the Column I with Column II and choose the correct option from the codes given below.

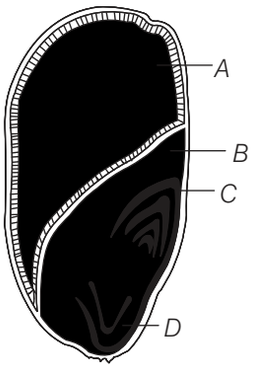
Column I	Column II
A. Phycophages	1. Fungal viruses
B. Mycophages	2. Carry enzyme reverse transcriptase
C. Retroviruses	3. Benign and malignant
D. Tumour viruses	4. Viruses which are parasitic on algae

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	3	2	1
(c)	4	1	2	3	(d)	1	3	2	4

94. K-T boundary refers to
(a) mass extinction of 60 million years ago when dinosaurs disappeared
(b) deposits of iridium which are rare on earth
(c) Both (a) and (b)
(d) None of these

95. Refer to the diagram given below and choose the option that indicates incorrect labelling.



(a) *A*–Represents the storage tissue rich in starch
(b) *B*–Represents scutellum
(c) *C*–Represents coleorhiza
(d) *D*–Represents the protective covering of radicle

96. Which of the following statements is/are incorrect?
I. The fast block polyspermy develops in response to the opening of sodium gates in the plasma membrane.
II. The slow block polyspermy develops in response to the release of bindin.
III. Capacitation is the process of entry of more than one sperm into the oocyte.

Choose the correct option.
(a) Only III (b) I and II
(c) Only I (d) I and III

97. Consider the following statements about the structure of ovule and select the one which is incorrect.
(a) The point of attachment of funicle with the body is called hilum
(b) A collar-shaped outgrowth from the base of the ovule which forms the third integument is called aril
(c) In a polygonum type of embryo sac, the egg apparatus is situated towards the chalazal end
(d) Generally, both the polar nuclei fuse before fertilisation to form a single diploid nucleus

98. Which of the following statements is true?
(a) Auxin translocation is non-polar
(b) Abscisic acid is transported through phloem only
(c) Phototropism is induced by blue light
(d) Auxin is not affected by TIBA

99. Which among the following is/are adaptations of hydrophytes?
(a) No cuticle over the epidermis
(b) Aerenchyma is well-developed
(c) Secondary growth is absent
(d) All of the above

100. The enzyme that catalyses the conversion of plasminogen to plasmin and is used in acute coronary arterial thrombosis is
(a) streptokinase (b) phosphotransferase
(c) β-lactamase (d) penicillinase

101. The production of fertile soil on bare rocks is
(a) secondary succession (b) primary succession
(c) heterotrophic succession (d) allogenic succession

102. The Tumour inducing (Ti) plasmid of which plant pathogen has been modified into a cloning vector to deliver genes of our interest?
(a) *Providencia stuartii* (b) *Agrobacterium tumefaciens*
(c) *Haemophilus haemolyticus* (d) *Bacillus globigii*

103. Which of the following disorders occurs due to the deficiency in immune system?
(a) Parkinson's disease (b) Digeorges' syndrome
(c) Huntingtons's chorea (d) Alzheimer's disease

104. In which of the following the percentage of lipoprotein is maximum?
(a) Chylomicron (b) VLDL
(c) DHL (d) LDL

105. Which among the following statements is/are incorrect regarding *Spirogyra*?
(a) Fertilisation takes place in the female gametangium
(b) Conjugation tube is not formed
(c) Both (a) and (b)
(d) None of the above

106. Which of the following is the intermediate compound linking glycolysis to Krebs cycle?
(a) Oxaloacetate (b) Acetyl Co-A
(c) Pyruvate (d) Citric acid

107. Catabolite Activator Protein (CAP) is /an
(a) inducer (b) corepressor
(c) aporepressor (d) apoinducer

108. What will happen if corpora allata is removed from the body of cockroach?
(a) Digestion will not occur (b) It will become sterile
(c) It will become adult (d) It will become blind

109. Identify the incorrectly matched pair.
(a) Physical xerophytes – Soils which are physically dry
(b) Physiological xerophytes – Soils having sufficient water
(c) Both (a) and (b)
(d) None of the above

110. *trp* operon is regulated by the availability of amino acid tryptophan, this mechanism is known as
(a) termination (b) attenuation
(c) suppression (d) repression

111. In 'protoplast fusion', tissues of two different plants are cultured. Their cells are separated and subjected to the treatment with
(a) IBA and NAA
(b) ethylene and glycol
(c) coconut milk and auxin
(d) pectinase and cellulase

112. Which are the first organs to develop in a human embryo?
(a) Heart and blood vessels (b) Eyes and ears
(c) Brain and spinal cord (d) Mouth and tongue

113. What would be the number of chromosomes in aleurone cells of a plant when number of chromosomes in leaf cell is 24?
(a) 12 (b) 24
(c) 36 (d) 48

114. Which among the following statements is/are incorrect regarding significance of biopatents?
(a) Permit private, monopoly rights over cells, genes, animals and plants
(b) People would not research in such areas, which are dominated by patents
(c) Lead to research programmes dominated by potentiability and profitability
(d) Philosophy and social commentary that deal with the biological sciences and their potential impact on society

115. Bordered pits are found on the radial walls of
(a) xylem tracheids (b) sieve tubes
(c) xylem fibres (d) sieve plates

116. In bacteria, glucose is utilised first, even if other sources of sugar are available. This happens through a mechanism known as
(a) catabolite repression (b) enzyme repression
(c) operon repression (d) positive feedback mechanisms

117. The life history traits of organisms have evolved in relation to
(a) Darwinian fitness
(b) organisms evolve under selection pressure developed by environmental factors
(c) organisms achieve most efficient reproductive strategy
(d) All of the above

118. In mitotic prophase, the sister chromatids are held together by a multi-subunit protein complex called
(a) actin (b) cytokinin (c) cohesin (d) tubulin

119. Viruses are known as wandering genes because
(a) nucleic acid is the only active part of a virus
(b) capsid is the outer protective coat
(c) nucleic acid is the central core
(d) capsid is made up of specific protein

120. A person performing 'Yoga' breathes in as much air as possible. The volume of air inspired is called
(a) tidal volume (b) inspiratory capacity
(c) vital capacity (d) inspiratory reserve volume

121. In which of the following tissues the benign tumour is enclosed?
(a) Connective tissue (b) Epithelial tissue
(c) Muscular tissue (d) Nervous tissue

122. From which amino acid, creatine is metabolically formed?
(a) Histidine (b) Phenylalanine
(c) Tryptophan (d) Arginine

123. *c*AMP regulates the *lac* operon by
(a) binding to the *lac* repressor
(b) binding to the operator
(c) combining with CAP to form a complex which starts transcription by binding near the promoter
(d) All of the above

124. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	CO_2	1.	Freshwater wetlands
B.	CH_4	2.	Non-toxic gas
C.	CFC	3.	Burning of fossil fuel
D.	N_2O	4.	Industrial processes

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	3	1	2	4
(c)	4	2	3	1	(d)	1	3	2	1

125. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Intrafascicular cambium	1.	Cambium between xylem and phloem
B.	Interfascicular cambium	2.	Parenchymatous cells of medullary rays
C.	Tyloses	3.	Ballon-like structures
D.	Heartwood	4.	Physiologically inactive

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	2	4	(b)	3	1	2	4
(c)	1	2	3	4	(d)	4	1	2	3

126. Which of the following is not a secondary sex organ in human males?

(a) Seminal vesicle (b) Penis
(c) Epididymis (d) Testis

127. With respect to eukaryotic gene regulation, which of the following statements is not true for enhancers?

(a) They function in tissue specific manner
(b) They function in any orientation
(c) They function as promoters
(d) They function even when are at a distance from the gene

128. Thyroxine is produced from which of the following amino acids?

(a) Tryptophan (b) Alanine (c) Glycine (d) Tyrosine

129. Identify the incorrect statement(s).

I. Ability of an organism to maintain a state of dynamic constancy is called thermoregulation.
II. The maintenance of constant internal temperature by an organism is called homeostasis.
III. The oldest living plant on earth is the great Indian banyan tree.

Choose the correct option.

(a) Only I (b) I and II (c) Only III (d) All of these

130. Castle's intrinsic factor is a glycoprotein secreted by parietal cells is required for the absorption of

(a) cobalamine (b) biotin (c) pyridoxine (d) niacin

131. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I (Type of bryophytes)		Column II (Features)
A.	Hepaticopsida	1.	Capsule without any columella
B.	Anthocerotopsida	2.	Sterile columella present
C.	Bryopsida	3.	Sporophyte well-differentiated into foot, seta and capsule
D.	Marchantiales	4.	*Riccia*

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	4	3	(b)	1	2	3	4
(c)	2	1	3	4	(d)	4	1	3	2

132. In golden rice *crtI* (carotene desaturase) gene has been inserted from the

(a) *Narcissus pseudonarcissus* (b) *Erwinia uredovora*
(c) *Pseudomonas putida* (d) *E. coli*

133. Match the following columns and choose the correct option from the codes given below.

	Column I (Hormone)		Column II (Function)
A.	Human chorionic gonadotropin	1.	Promotes growth of myometrium
B.	Progesterone	2.	Maintains corpus luteum
C.	Oestrogen	3.	Suppresses uterine contractions
D.	Relaxin	4.	Softens cervix to prepare for dilation

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	1	4	(b)	3	2	4	1
(c)	4	1	3	2	(d)	1	4	2	3

134. The individuals produced as a consequence of the undersecretion of pituitary gland are called

(a) midgets (b) fidgets (c) gadgets (d) maggots

135. In which of the following, males have half the number of chromosomes that females possess?

(a) Moth (b) Chicken (c) Honeybee (d) Pigeon

136. Which of the following statements is/are correct?

(a) Bilirubin is transported in blood by being attached to plasma albumin
(b) Bilirubin is insoluble in water
(c) Both (a) and (b) are correct
(d) None of the above

137. In which of the following is XX-XO sex chromosome complement found?
(a) Chimpanzee (b) Birds (c) Honeybee (d) Cockroach

138. The effect(s) of ozone depletion include(s)
(a) inhibition of photosynthesis in phytoplanktons
(b) damage in nucleic acids of living organisms
(c) increase in UV-radiations reaching the earth surface
(d) All of the above

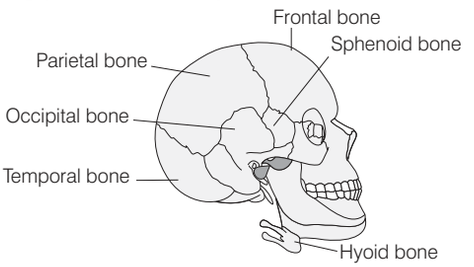
139. In which of the following structures insulin does not increase the glucose uptake?
(a) Cardiac muscles and skeletal muscles
(b) Renal tubules and intestinal mucosa
(c) Adipose tissues and cardiac muscles
(d) Smooth muscles and skeletal muscles

140. Which of the following is/are required for full expression of the *lac* operon?
(a) Lactose (b) Allolactose
(c) Lactose and *c*AMP (d) Allolactose and *c*AMP

141. The human biochemical disorder Tay-Sachs disease is an example of
(a) incomplete dominance (b) codominance
(c) epistasis (d) multiple alleles

142. Which among the following statements is/are correct regarding water potential?
(a) It refers to the chemical potential of water in a system
(b) Water potential of pure water is zero
(c) Water potential increases when pure water is heated
(d) All of the above

143. In the figure given below, a diagrammatic view of human skull is given. Two of the bones are wrongly labelled. They are



(a) sphenoid and hyoid bone (b) frontal and parietal bone
(c) occipital and temporal bone (d) parietal and occipital bone

144. Which of the following is a 'double-headed snake'?
(a) *Python* (b) Viper (c) Sand boa (d) Kraits

145. Which of the following is a pair of zoogeographical regions separated by high mountains?
(a) Oriental and Australian (b) Nearctic and Neotropical
(c) Ethiopian and Nearctic (d) Palaearctic and Oriental

146. A little boy was inquisitive about why a cow was licking its calf. What is the most probable explanation given by his grandfather?
(a) It is an action of time passing with calf
(b) Lysozymes present in the saliva of the mother protects the calf from bacterial infection
(c) To remove insects sitting on the calf
(d) Both (a) and (b)

147. Craidukov's phenomenon is
(a) the capacity of changing colour
(b) the change in colour according to the wavelength of incident light
(c) also known as chromatic adaptation
(d) All of the above

148. Which of the following statements are not disadvantages of hydroponics?
I. The cost of these experiments is high.
II. It requires skilled people.
III. It facilitates production of seasonal vegetables.
IV. It avoids soil borne pathogens.

Choose the correct option.
(a) II and III (b) III and IV (c) I and IV (d) I, II and III

149. Which of the following classes of Echinodermata does not include free-moving echinoderms?
(a) Asteroidea (b) Holothuroidea
(c) Echinoidea (d) Crinoidea

150. Identify the incorrect statement(s).
I. Amniotic fluid test is used for diagnosing chromosomal abnormalities.
II. In ultrasound imaging, visible images are produced by the pattern of echos made by organs.
III. Chronic villi sampling is an invasive technique used in prenatal diagnosis.
IV. Maternal blood sampling is an invasive technique where mother's blood is tested.

Choose the correct option.
(a) Only I (b) Only IV (c) Only II (d) III and IV

151. Pick out the mismatched pair(s).

I.	Wind	— Anemophily	—	Maize
II.	Water	— Hydrophily	—	*Vallisneria*
III.	Insects	— Entomophily	—	Poppy
IV.	Snails	— Myrmecophily	—	*Lemna*
V.	Ants	— Malacophily	—	*Chrysanthemum*

Choose the correct option.
(a) Only IV (b) Only V
(c) IV and V (d) None of these

152. Which among the following statements is incorrect regarding viruses?
(a) Viruses possess no membranes of their own
(b) Viruses do not possess ribosomes to make proteins
(c) Viruses do not have any source of energy (ATP)
(d) Viruses can reproduce by themselves

153. The primary treatment of effluents involves
(a) removal of large particles (b) removal of small particles
(c) removal of dissolved particles (d) Both (a) and (b)

154. On the basis of functions, the meristematic tissue is divided into
(a) protoderm, procambium, ground meristem
(b) apical, lateral, intercalary
(c) mass, plate, rib meristem
(d) primary, secondary

155. Identify the incorrectly matched pair.
(a) Chlorophyceae–Chlorophyll-*a* and *b*, carotenoids
(b) Phaeophyceae–Chlorophyll-*a* and *c*, fucoxanthin
(c) Rhodophyceae–Chlorophyll-*a*, phycocyanin, phycoerythrin, xanthophyll
(d) None of the above

156. The MN blood group in humans is an example of
(a) partial dominance (b) codominance
(c) incomplete dominance (d) epistasis

157. Which of the following category of genes found in normal cells have an antiproliferative function?
(a) Oncogenes (b) Proto-oncogenes
(c) Retro genes (d) Tumour-suppressor genes

158. In the glomerulus of nephron, the afferent arteriole is
(a) longer than efferent arteriole
(b) of same diameter as efferent arteriole
(c) wider than efferent arteriole
(d) narrower than efferent arteriole

159. Which among the following is/are method(s) of safe disposal of radioactive wastes?
(a) Radioactive waste is buried deep under ground surface
(b) Radioactive waste is reprocessed
(c) Radioactive waste is buried in specially constructed buildings
(d) All of the above

160. The oldest known fossil cells were about the size of modern prokaryotes. Which of the following resembles them?
(a) Red algae (b) Green algae (c) *Amoeba* (d) Bacteria

161. Which one among the following is not the characteristic of virusoids?
(a) They are small circular RNA
(b) They are always associated with larger viral RNA
(c) They may form part of the viral genome
(d) They are proteinaceous infectious particles

162. 'Saheli' is a new oral contraceptive pill for females with
(a) low side effects and low contraceptive value
(b) high side effects and high contraceptive value
(c) low side effects and high contraceptive value
(d) high side effects and low contraceptive value

163. MAB (Man And Biosphere) programme is associated with
(a) studying impact of human interference on environment
(b) pollution in biotic and abiotic environments
(c) conservation strategies for present and future
(d) All of the above

164. It is important to make a newborn baby burp immediately after feeding or else they may regurgitate. This is because of lesser developed
(a) cardiac sphincter (b) gastric cells
(c) pyloric sphincter (d) None of these

165. Identify the incorrectly matched pair.
(a) Sclerophyllous–Stiff leaves
(b) Trichophyllous–Leaves covered with hairs
(c) Malacophyllous–Soft and fleshy leaves
(d) None of the above

166. Clones can be obtained by which of the following techniques?
(a) Vegetative propagation (b) Cultivation
(c) Hybridisation (d) Sexual reproduction

167. A health conscious lady takes a lot of germinated lentils in breakfast. Which of the following enzymatic reaction will be the first to occur after its intake in mouth?

	Substrate	Enzyme	Products formed
(a)	Protein	Pepsin	Polypeptide
(b)	Fats	Lipase	Diglycerides
(c)	Maltose	Maltase	Glucose + Glucose
(d)	Starch	Amylase	Glucose + Fructose

168. Which of the following organs of the cockroach absorbs nitrogenous waste from haemolymph?
(a) Mycetocytes (b) Urate cells
(c) Utriculi majores (d) Anal cerci

169. Which among the following statements is/are true for carbon cycle?
(a) Atmospheric inputs of carbon through rainfall are higher
(b) There is respiratory release of carbon into the atmosphere
(c) The gaseous exchange of carbon between organisms and environment is negligible
(d) Both (a) and (b)

170. Real time PCR utilises ..... for detection and estimation of DNA products.
(a) methyl orange (b) SYBR green
(c) phenolphthalein (d) ethidium bromide

171. 'Brown sugar' is composed of
(a) lysergic acid diethylamide
(b) tetra-hydro cannibinols
(c) diacetyl morphine hydrochloride
(d) acetaldehyde

172. Match Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Tropical rainforests	1.	Sal
B.	Tropical deciduous forests	2.	*Acacia* sp.
C.	Desert	3.	*Dipterocarpus*
D.	Coastal biome	4.	Mangroves

Codes

	A	B	C	D		A	B	C	D
(a)	3	1	2	4	(b)	1	2	3	4
(c)	2	1	3	4	(d)	4	1	3	2

173. Match the contents of Column I to the contents of Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Barley	1.	Atomita-2
B.	Peppermint	2.	Aruna
C.	Castor	3.	Todd's Mitcham
D.	Rice	4.	Erectiferum
		5.	Pusa Lerma

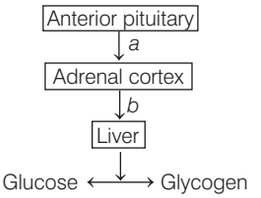
Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	3	2	1
(c)	3	4	1	2	(d)	2	1	3	4

174. Why the bronchioles and lungs in human beings do not collapse even though they lack the cartilaginous rings?

(a) Surfactants secreted by the clara cells prevent the collapse
(b) The intercostal muscles keep the pleura expanded
(c) The elastic cartilage present on their surface prevents them from collapsing
(d) None of the above

175. Refer to the following flowchart and select the option which correctly indicates '*a*' and '*b*'.



(a) CRF and ACTH
(b) ACTH and cortisol
(c) CRF and cAMP
(d) STH and cortisol

176. In *Drosophila*, gynandromorphs occur due to the

(a) loss of autosomal chromosome
(b) loss of X-chromosome
(c) gain of Y-chromosome
(d) chemical induction

177. Along with scales, scutes and bony plates are found in which of the following?

(a) *Torpedo*
(b) Salmon
(c) *Gambusia*
(d) *Hippocampus*

178. According to the 'Blackman's law of limiting factor', at a particular time, photosynthesis can be limited by

(a) CO_2 concentration
(b) light
(c) Both (a) and (b)
(d) Either (a) or (b)

179. Identify the incorrectly matched pair.

(a) Ebola → Filoviruses
(b) Mumps → Paramyxovirus
(c) Rabies → Rhabdovirus
(d) Yellow fever → Variola virus

180. The arrangement of outer and central microtubules in a cilium is called

(a) 9+2 pattern
(b) 9+0 pattern
(c) 8+2 pattern
(d) 9+1 pattern

